LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Physical properties, composition and division of atmosphere, International Standard Atmosphere, variation of properties of air with altitude
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Physical properties, composition and division of atmosphere, International Standard Atmosphere, variation of properties of air with altitude.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding atmospheric layers; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary physical properties of air?
(b) Name the four main layers of the atmosphere.
(c) Define the International Standard Atmosphere.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Composition of dry air • Physical properties of air • Flow over bodies • Boundary layer concept	L & Ds	PPT, IB	15 min
(b)	• Division of atmosphere • International Standard Atmosphere • ISA temperature lapse rate • ISA pressure variation	L & Ds	PPT, PDF	20 min
(c)	• Variation of density with altitude • Variation of temperature with altitude • ISA deviation effects • Standard sea level values	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how air density changes as altitude increases in the troposphere.
(b) Describe the significance of the boundary layer in aerodynamic flow.
(c) State the standard temperature and pressure values at sea level under ISA conditions.

12. Home Assignment and Follow Up	Prepare a brief note comparing geometric altitude, pressure altitude, and density altitude, referencing the concepts covered in the next lesson plan (LP 02).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Pressure altitude, density altitude, geometric, absolute altitude and standard altitude	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min